

UNIVERSITY OF BRISTOL

Faculty of Science

**MOCK EXAMINATION — Paper 8 (Extreme) —
SOLUTIONS AND MARK SCHEME****Core Physics I — Classical, Quantum and Thermal Physics**

Module Code: PHYS10012

Total Marks: 100

Marking Notes:

- Mark allocations are shown in **blue** at each step.
 - Final answers are highlighted in green boxes.
 - “*(From formula sheet)*” indicates the equation is on the provided formula sheet.
 - “*(Recalled — not on formula sheet)*” indicates the equation must be known from memory.
 - Award partial marks for correct method even if the final answer is wrong.
 - Accept equivalent correct forms of answers.
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Section A: Oscillations and Waves — Answers

1. **A** — For small x , use the Taylor expansion $\cos(x/a) \approx 1 - \frac{1}{2}(x/a)^2$, so $V(x) \approx V_0 \cdot \frac{1}{2}(x/a)^2 = \frac{V_0}{2a^2} x^2$. The restoring force is $F = -dV/dx = -(V_0/a^2)x$ (*From formula sheet*). Comparing with $F = -kx$, the effective spring constant is $k_{\text{eff}} = V_0/a^2$, so $\omega = \sqrt{k_{\text{eff}}/m} = \sqrt{V_0/(ma^2)}$.
2. **B** — Since energy is proportional to amplitude squared, $E(t) = \frac{1}{2}kA(t)^2 \propto A^2$. With $A(t) = A_0 e^{-\gamma t/2}$, we get $E(t) = E_0 e^{-\gamma t}$.
3. **B** — For a lightly damped oscillator ($\gamma \ll \omega_0$, so $Q \gg 1$), the power absorption curve is a Lorentzian centred at ω_0 . The FWHM is $\Delta\omega = \gamma$. Note that $\gamma = \omega_0/Q$ (*From formula sheet*), so options B and D are equivalent.
4. **C** — When $A_1 \neq A_2$, the superposition can be decomposed as a standing wave with amplitude $\min(A_1, A_2)$ plus a travelling wave with amplitude $|A_1 - A_2|$. The result is *not* a pure standing wave (which requires equal amplitudes) nor a pure travelling wave.
5. **C** — A quarter-wave transformer eliminates reflection when the impedance of the matching section satisfies $Z_m = \sqrt{Z_1 Z_3}$, the geometric mean of the two outer impedances. This is the impedance analogue of quarter-wave anti-reflection coatings in optics.
6. **A** — For mass 1: $m\ddot{x}_1 = -kx_1 - k_c(x_1 - x_2) = -(k + k_c)x_1 + k_c x_2$. For mass 2: $m\ddot{x}_2 = -kx_2 - k_c(x_2 - x_1) = k_c x_1 - (k + k_c)x_2$. In matrix form this gives $\mathbf{K} = \begin{pmatrix} k + k_c & -k_c \\ -k_c & k + k_c \end{pmatrix}$.
7. **B** — Dispersion (spreading of a wave packet) occurs when different frequency components travel at different group velocities, which happens when the group velocity varies with k , i.e. $dv_g/dk = d^2\omega/dk^2 \neq 0$. A non-zero first derivative $d\omega/dk$ merely means the packet moves; it does not cause spreading.
8. **D** — Expanding to first order in β : the classical result gives $f'_{\text{cl}} \approx f_0(1 + \beta)$. The relativistic result is $f'_{\text{rel}} = f_0 \sqrt{(1 + \beta)/(1 - \beta)} \approx f_0(1 + \beta/2)(1 + \beta/2) = f_0(1 + \beta)$ to first order. Both agree at order β ; the difference appears only at order β^2 (the transverse Doppler effect term $\beta^2/2$).
9. **A** — The standing wave $y = 2A \sin(kx) \cos(\omega t)$ has maximum amplitude $2A$, while the travelling wave has amplitude A . At the instant of maximum displacement, all energy is potential. The energy density of the standing wave scales as $(2A)^2$ while the travelling wave scales as A^2 , but averaged over a wavelength, the standing wave energy per unit length is twice that of the single travelling wave. (This is expected since the standing wave is the superposition of two equal-amplitude counter-propagating waves.)
10. **C** — The Mach cone half-angle satisfies $\sin \theta = v_{\text{sound}}/v_{\text{object}} = 1/\text{Ma} = 1/2$, so $\theta = 30^\circ$.

Section B: Quantum Physics — Answers

1. **B** — By the antisymmetry of the commutator: $[\hat{B}, \hat{A}] = \hat{B}\hat{A} - \hat{A}\hat{B} = -(\hat{A}\hat{B} - \hat{B}\hat{A}) = -[\hat{A}, \hat{B}] = -i\hat{C}$.
2. **B** — If $[\hat{A}, \hat{B}] = 0$, the two operators are said to be compatible. They possess a complete set of simultaneous eigenstates, so both observables can be measured simultaneously with no mutual disturbance. This does *not* mean $\hat{A} = \hat{B}$, nor that their measurement outcomes are equal.
3. **C** — If U is the unitary matrix of eigenvectors of H , then $U^\dagger H U = D$, where D is the diagonal matrix of eigenvalues. This is the standard spectral decomposition / unitary diagonalisation of a Hermitian matrix.
4. **C** — The no-cloning theorem follows from the linearity of quantum mechanics. If a cloning unitary U could copy any state, then applying it to a superposition would (by linearity) produce a different result from copying each basis state individually, leading to a contradiction. The key ingredient is linearity (unitarity), not the uncertainty principle.
5. **C** — For a pure state, $\rho = |\psi\rangle\langle\psi|$ satisfies $\rho^2 = \rho$ (it is a projector). Therefore $\text{Tr}(\rho^2) = \text{Tr}(\rho) = 1$. (For a mixed state, $\text{Tr}(\rho^2) < 1$, which is the purity criterion.)
6. **B** — A superposition $c_1 |E_1\rangle + c_2 |E_2\rangle$ evolves with time-dependent phases $e^{-iE_1 t/\hbar}$ and $e^{-iE_2 t/\hbar}$ (*From formula sheet*). Measurement probabilities oscillate with angular frequency $\omega = |E_1 - E_2|/\hbar$, giving a characteristic timescale $\tau = 1/\omega = \hbar/|E_1 - E_2|$.
7. **B** — $S_{1z}S_{2z} |\uparrow\downarrow\rangle = (+\hbar/2)(-\hbar/2) |\uparrow\downarrow\rangle = -(\hbar^2/4) |\uparrow\downarrow\rangle$ and similarly $S_{1z}S_{2z} |\downarrow\uparrow\rangle = (-\hbar/2)(+\hbar/2) |\downarrow\uparrow\rangle = -(\hbar^2/4) |\downarrow\uparrow\rangle$. Therefore $\langle S_{1z}S_{2z} \rangle = \frac{1}{2}(-\hbar^2/4) + \frac{1}{2}(-\hbar^2/4) = -\hbar^2/4$.
8. **B** — $\rho = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle)(\langle\uparrow\uparrow| + \langle\downarrow\downarrow|)$. Taking the partial trace over qubit 2: $\rho_1 = \text{Tr}_2(\rho) = \frac{1}{2}(\langle\uparrow_2|\uparrow_2\rangle |\uparrow_1\rangle\langle\uparrow_1| + \langle\downarrow_2|\downarrow_2\rangle |\downarrow_1\rangle\langle\downarrow_1|) = \frac{1}{2}(|\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow|) = \frac{1}{2}\hat{I}$, the maximally mixed state.
9. **C** — With 6 fermions and 2 per level, we fill levels $n = 1, 2, 3$. The energies are $2(1^2) + 2(2^2) + 2(3^2) = 2 + 8 + 18 = 28$ in units of E_1 . So the ground-state energy is $28 E_1$.
10. **B** — Parity (P) is violated in weak interactions — this was demonstrated by Wu's experiment (1957) on ^{60}Co beta decay. Electromagnetic and strong interactions conserve parity. The combined symmetry CPT is conserved in all known interactions.

Section C: Long Answer Solutions

Question 1.

[15 marks]

Snooker Ball Impulse, Rolling Friction, and Loop-the-Loop

- (a) (i) Applying the impulse–momentum theorem for translation: (*Recalled — not on formula sheet*)

$$J = Mv_0 \implies v_0 = \frac{J}{M} \quad [1]$$

For rotation, the impulse J acts at a lever arm h from the centre of mass, producing a torque impulse about the CM:

$$Jh = I\omega_0$$

For a solid sphere, $I = \frac{2}{5}MR^2$ (*From formula sheet*), so:

$$Jh = \frac{2}{5}MR^2\omega_0 \implies \omega_0 = \frac{5Jh}{2MR^2} = \frac{5v_0h}{2R^2}$$

$$v_0 = J/M, \quad \omega_0 = 5Jh/(2MR^2)$$

[1]

- (ii) The rolling-without-slipping condition is $v_0 = R\omega_0$: (*Recalled — not on formula sheet*)

$$\frac{J}{M} = R \cdot \frac{5Jh}{2MR^2} = \frac{5Jh}{2MR} \quad [1]$$

Dividing both sides by J/M :

$$1 = \frac{5h}{2R} \implies h = \frac{2R}{5} \quad [1]$$

The height above the table is the height above the lowest point, which is the centre height R plus h :

$$h = \frac{2}{5}R \text{ above the centre, i.e. the cue should strike at height } R + \frac{2}{5}R = \frac{7}{5}R \text{ above the table.}$$

[1]

- (b) (i) During sliding, kinetic friction $f = \mu_k Mg$ acts backward (opposing the forward sliding motion): (*Recalled — not on formula sheet*)

$$\text{Translation: } Ma = -\mu_k Mg \implies a = -\mu_k g \quad [1]$$

Rotation (friction torque accelerates rotation): $\tau = fR = \mu_k MgR$

$$I\alpha = \mu_k MgR \implies \frac{2}{5}MR^2\alpha = \mu_k MgR$$

$$a = -\mu_k g \text{ (deceleration),} \quad \alpha = \frac{5\mu_k g}{2R} \text{ (angular acceleration)}$$

[1]

- (ii) The velocities during sliding are:

$$v(t) = v_0 - \mu_k g t$$

$$\omega(t) = \omega_0 + \frac{5\mu_k g}{2R} t$$

Rolling begins when $v = R\omega$:

$$v_0 - \mu_k g t = R \left(\omega_0 + \frac{5\mu_k g}{2R} t \right) = R\omega_0 + \frac{5\mu_k g}{2} t$$

$$v_0 - R\omega_0 = \mu_k g t + \frac{5\mu_k g}{2} t = \frac{7\mu_k g}{2} t$$

$$t_{\text{roll}} = \frac{2(v_0 - R\omega_0)}{7\mu_k g}$$

[1]

For a centre-strike, $h = 0$ so $\omega_0 = 0$:

$$t_{\text{roll}} = \frac{2v_0}{7\mu_k g}$$

$$v_{\text{roll}} = v_0 - \mu_k g \cdot \frac{2v_0}{7\mu_k g} = v_0 - \frac{2v_0}{7} = v_0 \left(1 - \frac{2}{7} \right)$$

$$v_{\text{roll}} = \frac{5}{7} v_0$$

[1]

- (c) At the top of the loop, for minimum contact the normal force $N = 0$, so gravity alone provides the centripetal force: (*Recalled — not on formula sheet*)

$$Mg = \frac{Mv_{\text{top}}^2}{r}$$

$$v_{\text{top}}^2 = gr = 9.81 \times 0.8 = 7.848 \text{ m}^2/\text{s}^2$$

Using energy conservation from height H to the top of the loop (height $2r$). The disk rolls without slipping, so the total KE includes rotational energy. For a solid disk, $I = \frac{1}{2}MR^2$ (*From formula sheet*):

$$MgH = Mg(2r) + \frac{1}{2}Mv_{\text{top}}^2 + \frac{1}{2}I\omega_{\text{top}}^2$$

With rolling condition $v = R\omega$ and $I = \frac{1}{2}MR^2$:

$$\frac{1}{2}I\omega^2 = \frac{1}{2} \cdot \frac{1}{2}MR^2 \cdot \frac{v^2}{R^2} = \frac{1}{4}Mv^2$$

$$\text{So total KE} = \frac{1}{2}Mv^2 + \frac{1}{4}Mv^2 = \frac{3}{4}Mv^2.$$

$$MgH = Mg(2r) + \frac{3}{4}Mv_{\text{top}}^2$$

$$gH = 2gr + \frac{3}{4}v_{\text{top}}^2 = 2gr + \frac{3}{4}gr = \frac{11}{4}gr$$

$$H = \frac{11r}{4} = \frac{11 \times 0.8}{4}$$

$$H = 2.2 \text{ m}$$

[1]

Question 2.

[15 marks]

Advanced Entropy, Lennard-Jones Potential, and Maxwell–Boltzmann Distribution

- (a) (i) For a reversible (Carnot) process between two finite thermal reservoirs, at each infinitesimal step a Carnot engine operates between temperatures T_h (hot block) and T_c (cold block): [1]

The hot block loses heat $dQ_h = -mc dT_h$ (it cools), and the cold block gains heat $dQ_c = mc dT_c$ (it warms). For a Carnot engine:

$$\frac{dQ_c}{dQ_h} = \frac{T_c}{T_h}$$

$$\frac{mc dT_c}{-mc dT_h} = \frac{T_c}{T_h} \implies \frac{dT_c}{T_c} = -\frac{dT_h}{T_h}$$

Integrating from initial to final states:

$$\ln\left(\frac{T_f}{T_2}\right) = -\ln\left(\frac{T_f}{T_1}\right) = \ln\left(\frac{T_1}{T_f}\right)$$

$$\frac{T_f}{T_2} = \frac{T_1}{T_f} \implies T_f^2 = T_1 T_2$$

$$T_f = \sqrt{T_1 T_2}$$

[1]

- (ii) The total entropy change of the two blocks: (*From formula sheet*)

$$\Delta S = mc \ln\left(\frac{T_f}{T_1}\right) + mc \ln\left(\frac{T_f}{T_2}\right) = mc \ln\left(\frac{T_f^2}{T_1 T_2}\right)$$

Since $T_f^2 = T_1 T_2$:

$$\Delta S = mc \ln\left(\frac{T_1 T_2}{T_1 T_2}\right) = mc \ln(1)$$

$$\Delta S_{\text{total}} = 0, \text{ confirming the process is reversible.}$$

[2]

- (b) (i) By conservation of energy with no work done: (*Recalled — not on formula sheet*)

$$mc T_1 + mc T_2 = 2mc T'_f$$

$$T'_f = \frac{T_1 + T_2}{2}$$

[1]

- (ii) The total entropy change is: (*From formula sheet*)

$$\Delta S_{\text{total}} = mc \ln\left(\frac{T'_f}{T_1}\right) + mc \ln\left(\frac{T'_f}{T_2}\right) = mc \ln\left(\frac{(T'_f)^2}{T_1 T_2}\right) = mc \ln\left(\frac{(T_1 + T_2)^2}{4T_1 T_2}\right) \quad [1]$$

By the AM–GM inequality, $(T_1 + T_2)/2 \geq \sqrt{T_1 T_2}$, with equality only when $T_1 = T_2$. Therefore $(T_1 + T_2)^2/(4T_1 T_2) \geq 1$, so $\Delta S_{\text{total}} \geq 0$.

$$\Delta S_{\text{total}} = mc \ln \left(\frac{(T_1 + T_2)^2}{4T_1 T_2} \right) > 0 \text{ for } T_1 \neq T_2.$$

[1]

(iii) In the reversible process, the initial total thermal energy is $mc(T_1 + T_2)$ and the final thermal energy of both blocks is $2mc\sqrt{T_1 T_2}$. The difference is work extracted:

$$W_{\text{max}} = mc(T_1 + T_2) - 2mc\sqrt{T_1 T_2} = mc(T_1 + T_2 - 2\sqrt{T_1 T_2}) = mc(\sqrt{T_1} - \sqrt{T_2})^2$$

[1]

(c) (i) Setting $dV/dr = 0$: (From formula sheet)

$$\frac{dV}{dr} = 4\varepsilon \left[-12 \frac{a^{12}}{r^{13}} + 6 \frac{a^6}{r^7} \right] = 0$$

$$12 \frac{a^{12}}{r^{13}} = 6 \frac{a^6}{r^7} \implies 2a^6 = r^6$$

[1]

$$r_0 = 2^{1/6} a$$

$$\text{At } r = r_0: V(r_0) = 4\varepsilon \left[\left(\frac{a}{2^{1/6}a} \right)^{12} - \left(\frac{a}{2^{1/6}a} \right)^6 \right] = 4\varepsilon \left[\frac{1}{2^2} - \frac{1}{2} \right] = 4\varepsilon \left[\frac{1}{4} - \frac{1}{2} \right] = 4\varepsilon \left(-\frac{1}{4} \right)$$

$$r_0 = 2^{1/6} a, \quad V(r_0) = -\varepsilon \text{ (well depth is } \varepsilon \text{)}.$$

[1]

(ii) Computing $V''(r)$:

$$V'(r) = 4\varepsilon [-12a^{12}r^{-13} + 6a^6r^{-7}]$$

$$V''(r) = 4\varepsilon [156a^{12}r^{-14} - 42a^6r^{-8}]$$

At $r_0 = 2^{1/6}a$:

$$\begin{aligned} V''(r_0) &= 4\varepsilon \left[\frac{156a^{12}}{(2^{1/6}a)^{14}} - \frac{42a^6}{(2^{1/6}a)^8} \right] = \frac{4\varepsilon}{a^2} \left[\frac{156}{2^{14/6}} - \frac{42}{2^{8/6}} \right] \\ &= \frac{4\varepsilon}{a^2} \left[\frac{156}{2^{7/3}} - \frac{42}{2^{4/3}} \right] = \frac{4\varepsilon}{a^2} \left[\frac{156}{4 \cdot 2^{1/3}} - \frac{42}{2 \cdot 2^{1/3}} \right] = \frac{4\varepsilon}{a^2 \cdot 2^{1/3}} \left[\frac{156}{4} - \frac{42}{2} \right] \\ &= \frac{4\varepsilon}{a^2 \cdot 2^{1/3}} [39 - 21] = \frac{4\varepsilon \times 18}{a^2 \cdot 2^{1/3}} \end{aligned}$$

$$k = V''(r_0) = \frac{72\varepsilon}{2^{1/3}a^2} = \frac{36 \times 2^{2/3} \varepsilon}{a^2}$$

[1]

Numerically:

$$k = \frac{72 \times 1.67 \times 10^{-21}}{1.2599 \times (3.40 \times 10^{-10})^2} = \frac{1.202 \times 10^{-19}}{1.2599 \times 1.156 \times 10^{-19}} = \frac{1.202 \times 10^{-19}}{1.456 \times 10^{-19}} = 0.826 \text{ N/m}$$

$$\omega = \sqrt{\frac{k}{m_{\text{Ar}}}} = \sqrt{\frac{0.826}{6.63 \times 10^{-26}}} = \sqrt{1.245 \times 10^{25}} = 3.53 \times 10^{12} \text{ rad/s}$$

$$f = \frac{\omega}{2\pi} = \frac{3.53 \times 10^{12}}{2\pi}$$

$$f \approx 5.6 \times 10^{11} \text{ Hz} \approx 0.56 \text{ THz}$$

[1]

This is in the correct range for atomic vibrations (far-infrared/THz range).

(d) Setting $c_{\text{mp}} = v_e$: (*From formula sheet*)

$$\sqrt{\frac{2k_B T}{m}} = v_e \implies T = \frac{mv_e^2}{2k_B} \quad [1]$$

$$T = \frac{6.64 \times 10^{-27} \times (11,200)^2}{2 \times 1.381 \times 10^{-23}} = \frac{6.64 \times 10^{-27} \times 1.254 \times 10^8}{2.762 \times 10^{-23}} \\ = \frac{8.33 \times 10^{-19}}{2.762 \times 10^{-23}} \quad [1]$$

$$T \approx 3.0 \times 10^4 \text{ K} = 30,000 \text{ K}$$

At room temperature ($T \approx 300 \text{ K}$), the most probable speed of helium is far below the escape velocity ($c_{\text{mp}} \approx 1,260 \text{ m/s}$ vs $v_e = 11,200 \text{ m/s}$). However, the Maxwell–Boltzmann distribution has a long high-energy tail extending well beyond c_{mp} . Over geological timescales, helium atoms in the tail of the distribution can exceed v_e and escape. This is why the Earth’s atmosphere contains very little helium despite it being common in the universe.

$T \approx 30,000 \text{ K}$. Helium *can* escape at room temperature via the high-energy tail of the MB distribution, over geological time.

[1]

Question 3.

[15 marks]

Advanced Boundary Conditions, Beaded String, and Wave Packets

- (a) (i) At $x = L$, the ring is massless and free to slide vertically, so there is no transverse restoring force. The vertical component of the string tension must vanish, which requires: [1]

$$\left. \frac{\partial y}{\partial x} \right|_{x=L} = 0 \quad (\text{a “free” or Neumann boundary condition — } x = L \text{ is an antinode}).$$

- (ii) The standing wave on a string fixed at $x = 0$ has the form:

$$y(x, t) = A \sin(kx) \cos(\omega t) \quad (\text{Recalled — not on formula sheet})$$

At $x = 0$: $\sin(0) = 0$ — satisfied automatically (node). [1]

$$\text{At } x = L: \left. \frac{\partial y}{\partial x} \right|_{x=L} = Ak \cos(kL) \cos(\omega t) = 0 \text{ for all } t, \text{ so } \cos(kL) = 0.$$

$$kL = \frac{(2n-1)\pi}{2}, \quad n = 1, 2, 3, \dots$$

$$k_n = \frac{(2n-1)\pi}{2L}, \quad \lambda_n = \frac{2\pi}{k_n} = \frac{4L}{2n-1}$$

$$\lambda_n = \frac{4L}{2n-1}, \quad n = 1, 2, 3, \dots$$

[1]

- (iii) The resonant frequencies are:

$$f_n = \frac{v}{\lambda_n} = \frac{v(2n-1)}{4L} = \frac{100 \times (2n-1)}{4 \times 2} = 12.5 (2n-1) \text{ Hz} \quad [1]$$

$$f_1 = 12.5 \text{ Hz}, \quad f_2 = 37.5 \text{ Hz}, \quad f_3 = 62.5 \text{ Hz}$$

[1]

- (b) (i) Boundary conditions at the walls: [1]

$$y(-L, t) = 0 \quad \text{and} \quad y(+L, t) = 0 \quad (\text{fixed ends})$$

$$\text{At } x = 0, \text{ the displacement must be continuous: } y(0^-, t) = y(0^+, t) \quad [1]$$

Force balance on the bead at $x = 0$ (the bead has mass m): The net transverse force on the bead equals $m\ddot{y}(0, t)$. The transverse force from the string tension is $T \partial y / \partial x$:

$$T \left. \frac{\partial y}{\partial x} \right|_{0^+} - T \left. \frac{\partial y}{\partial x} \right|_{0^-} = m \frac{\partial^2 y(0, t)}{\partial t^2}$$

[1]

(The discontinuity in slope is balanced by the bead's inertial force.)

- (ii) For symmetric (antisymmetric displacement) modes, $y(0, t) = 0$, meaning the bead is at a node and does not move. Then $\ddot{y}(0, t) = 0$, and the bead's inertia is irrelevant. [1]

Each half of the system is independently a string of length L fixed at both ends. The normal mode condition is:

$$kL = n\pi, \quad n = 1, 2, 3, \dots$$

$$\lambda_n = \frac{2L}{n}, \quad f_n = \frac{v}{\lambda_n} = \frac{nv}{2L} \text{ (From formula sheet)}$$

$$f_n = \frac{nv}{2L}, \quad n = 1, 2, 3, \dots \text{ (same as a single string of length } L \text{ fixed at both ends).}$$

[1]

(c) (i) Using $\omega = \alpha k^2$ with $\alpha = 0.5 \text{ m}^2/\text{s}$:

[1]

$$\omega_1 = 0.5 \times 10^2 = 50 \text{ rad/s}$$

$$\omega_2 = 0.5 \times 12^2 = 0.5 \times 144 = 72 \text{ rad/s}$$

$$\omega_1 = 50 \text{ rad/s}, \quad \omega_2 = 72 \text{ rad/s}$$

(ii) Average wavenumber: $\bar{k} = (10 + 12)/2 = 11 \text{ rad/m}$.

Phase velocity at $\bar{k}$: (Recalled — not on formula sheet)

$$v_p = \frac{\omega(\bar{k})}{\bar{k}} = \frac{\alpha \bar{k}^2}{\bar{k}} = \alpha \bar{k} = 0.5 \times 11 = 5.5 \text{ m/s}$$

[1]

Group velocity at $\bar{k}$: (From formula sheet)

$$v_g = \left. \frac{d\omega}{dk} \right|_{\bar{k}} = 2\alpha \bar{k} = 2 \times 0.5 \times 11 = 11 \text{ m/s}$$

[1]

$$v_p = 5.5 \text{ m/s}, \quad v_g = 11 \text{ m/s}$$

Note that $v_g = 2v_p$ for this quadratic dispersion relation.

(iii) Using the trig identity $\cos \alpha + \cos \beta = 2 \cos\left(\frac{\alpha+\beta}{2}\right) \cos\left(\frac{\alpha-\beta}{2}\right)$ (From formula sheet):

Let $\Delta k = k_2 - k_1 = 2 \text{ rad/m}$ and $\Delta \omega = \omega_2 - \omega_1 = 22 \text{ rad/s}$.

[1]

$$y(x, t) = 2A \cos\left(\frac{\Delta k}{2} x - \frac{\Delta \omega}{2} t\right) \cos(\bar{k} x - \bar{\omega} t)$$

where $\bar{\omega} = (\omega_1 + \omega_2)/2 = 61 \text{ rad/s}$.

$$y(x, t) = \underbrace{2A \cos(x - 11t)}_{\text{envelope (slow)}} \times \underbrace{\cos(11x - 61t)}_{\text{carrier (fast)}}$$

The envelope moves at speed $\Delta \omega / \Delta k = 22/2 = 11 \text{ m/s} = v_g$ (the group velocity).

The carrier moves at speed $\bar{\omega} / \bar{k} = 61/11 \approx 5.5 \text{ m/s} = v_p$ (the phase velocity).

[1]

Question 4.

[15 marks]

Three-State System, Fermion States, and Pion Decay(a) (i) Writing H in the ordered basis $\{|a\rangle, |b\rangle, |c\rangle\}$:

[1]

The diagonal elements are $\langle a|H|a\rangle = E_0$, $\langle b|H|b\rangle = E_0$, $\langle c|H|c\rangle = E_0$.The off-diagonal elements: $\langle a|H|b\rangle = V$, $\langle b|H|a\rangle = V$, and all others are zero.

$$H = \begin{pmatrix} E_0 & V & 0 \\ V & E_0 & 0 \\ 0 & 0 & E_0 \end{pmatrix}$$

(ii) The state $|c\rangle$ decouples completely (the third row and column have no off-diagonal elements coupling to $|a\rangle$ or $|b\rangle$). [1]Eigenvalue 1: E_0 , eigenstate $|c\rangle$.For the $\{|a\rangle, |b\rangle\}$ subspace, diagonalise the 2×2 block:

$$\begin{pmatrix} E_0 & V \\ V & E_0 \end{pmatrix}$$

The eigenvalues satisfy $(E_0 - \lambda)^2 - V^2 = 0$, giving $\lambda = E_0 \pm V$. [1]For $\lambda = E_0 + V$: eigenstate $|+\rangle = \frac{1}{\sqrt{2}}(|a\rangle + |b\rangle)$.For $\lambda = E_0 - V$: eigenstate $|-\rangle = \frac{1}{\sqrt{2}}(|a\rangle - |b\rangle)$.Eigenvalues: $E_0 + V$, $E_0 - V$, E_0 .Eigenstates: $\frac{1}{\sqrt{2}}(|a\rangle + |b\rangle)$, $\frac{1}{\sqrt{2}}(|a\rangle - |b\rangle)$, $|c\rangle$. [1](b) (i) Express the initial state $|a\rangle$ in the energy eigenbasis:

$$|a\rangle = \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle$$

(This follows from inverting $|\pm\rangle = (|a\rangle \pm |b\rangle)/\sqrt{2}$.)Applying time evolution (*From formula sheet*): [1]

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}e^{-i(E_0+V)t/\hbar}|+\rangle + \frac{1}{\sqrt{2}}e^{-i(E_0-V)t/\hbar}|-\rangle$$

Factoring out the common phase $e^{-iE_0t/\hbar}$:

$$|\psi(t)\rangle = e^{-iE_0t/\hbar} \left[\frac{1}{\sqrt{2}}e^{-iVt/\hbar}|+\rangle + \frac{1}{\sqrt{2}}e^{+iVt/\hbar}|-\rangle \right]$$

[1]

(ii) The probability amplitude to find the system in $|a\rangle$ is:

$$\langle a|\psi(t)\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}e^{-i(E_0+V)t/\hbar} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}e^{-i(E_0-V)t/\hbar}$$

[1]

(using $\langle a|+\rangle = 1/\sqrt{2}$ and $\langle a|-\rangle = 1/\sqrt{2}$)

$$= \frac{1}{2}e^{-iE_0t/\hbar}(e^{-iVt/\hbar} + e^{+iVt/\hbar}) = e^{-iE_0t/\hbar} \cos\left(\frac{Vt}{\hbar}\right)$$

$$P(a, t) = |\langle a | \psi(t) \rangle|^2 = \cos^2\left(\frac{Vt}{\hbar}\right)$$

[1]

The system oscillates (“Rabi oscillation”) between states $|a\rangle$ and $|b\rangle$ with angular frequency $2V/\hbar$.

- (c) Two identical spin- $\frac{1}{2}$ fermions require a total state (spatial $\otimes$ spin) that is antisymmetric under particle exchange. (*Recalled — not on formula sheet*) [1]

Spatial wavefunctions:

Symmetric (3 states): $|\alpha\alpha\rangle, |\beta\beta\rangle, \frac{1}{\sqrt{2}}(|\alpha\beta\rangle + |\beta\alpha\rangle)$

Antisymmetric (1 state): $\frac{1}{\sqrt{2}}(|\alpha\beta\rangle - |\beta\alpha\rangle)$

Spin wavefunctions:

Symmetric (triplet) (3 states): $|\uparrow\uparrow\rangle, \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), |\downarrow\downarrow\rangle$

Antisymmetric (singlet) (1 state): $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ [1]

Allowed combinations (antisymmetric total = symmetric $\otimes$ antisymmetric + antisymmetric $\otimes$ symmetric)

- 3 symmetric spatial $\times$ 1 antisymmetric spin (singlet) = 3 states
- 1 antisymmetric spatial $\times$ 3 symmetric spin (triplet) = 3 states

Total: $3 + 3 = 6$ antisymmetric two-particle states.

[1]

- (d) (i) The π^0 meson is a bound state of a quark and antiquark ($u\bar{u}$ or $d\bar{d}$ superposition). Its quantum numbers are: (*Recalled — not on formula sheet*)

	Charge	Baryon number	Lepton number
π^0	0	0	0
$\gamma + \gamma$	$0 + 0 = 0$	$0 + 0 = 0$	$0 + 0 = 0$

Charge: $0 = 0$ ✓. Baryon number: $0 = 0$ ✓. Lepton number: $0 = 0$ ✓.
All conserved.

[2]

- (ii) The π^0 has spin $J = 0$, so the total angular momentum of the two-photon system must be zero. [1]

In the centre-of-mass frame, the two photons travel in opposite directions (say along the z -axis). Each photon has spin 1 but, being massless, only has helicity $h = \pm 1$ (right-handed or left-handed circular polarisation). There is no $m = 0$ component for a photon.

For the photon travelling in the $+z$ direction, $J_z = h_1 = \pm 1$. For the photon travelling in the $-z$ direction, its angular momentum projected onto $+z$ is $-h_2$ (since its momentum is reversed).

The total $J_z = h_1 - h_2$. For $J_z = 0$, we need $h_1 = h_2$. Both photons must have the same helicity (both right-handed or both left-handed in their respective directions of travel). Equivalently, relative to a fixed axis, they have *opposite* helicities.

The $J = 0$ requirement is automatically satisfied since only $J_z = 0$ contributes and the overall angular momentum state is a scalar.

The two photons must have the same helicity (both RH or both LH). The decay is mediated by the **electromagnetic** force (involving photon emission from the quark-antiquark annihilation, described by the axial anomaly in QFT).

[1]